DIVYASHREE TARAFDER

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Currently in Manchester, UK · Open to relocation · UK Graduate Route visa valid until January 2029

PROFESSIONAL SUMMARY

Electrical power systems engineer with three years at Schneider Electric owning a low/medium-voltage automatic transfer switching and switchgear portfolio end-to-end, completing an MSc in Electrical Power Systems Engineering at the University of Manchester (2026). Combines power system analysis: load flow, short-circuit/fault level, protection coordination, transient stability and electromagnetic transient (EMT) modelling, with proven commercial and cross-functional delivery (P&L ownership, product roadmaps, pricing strategy. High-voltage plant exposure from transformer manufacturing and HV testing, and two peer-reviewed publications in power engineering. Seeking power systems, transmission & distribution, grid connection, consulting and engineering-management roles in the UK and India.

CORE SKILLS

Power Systems Studies and Analysis: Load flow / power flow, short-circuit & fault level analysis, protection grading & coordination, transient stability, electromagnetic transient (EMT) analysis, harmonics, contingency (N-1) analysis, economic dispatch & security-constrained operation, insulation coordination, reactive power / voltage control, OHL design, DER / rooftop-PV integration and network hosting-capacity assessment, IEC standards
Software & Tools: DIgSILENT PowerFactory, PSCAD, IPSA, MATLAB/Simulink, AutoCAD, SCADA, NI LabVIEW, PSpice, TINA, Proteus
Programming: Python (Jupyter), C++, Java, JavaScript, PLC
Professional: P&L management, offer strategy & product roadmaps, competitive pricing, technoeconomic analysis, demand forecasting, cross-functional team leadership, technical report writing, client & stakeholder presentations.

PROFESSIONAL EXPERIENCE

Deputy Offer Manager, Power Products (Automatic Transfer Switches & Switchgear)
Schneider Electric · Bengaluru, India · August 2022 - August 2025

- Owned the medium-voltage automatic transfer switch portfolio end-to-end, engaging utility and industrial customers and coordinating cross-functional teams (R&D, sales, supply chain, marketing), growing revenue 30% and improving gross margin 12% in FY2022-23.
- Led the localization of the ASCO 7000 Series IEC medium-voltage ATS under the “Make-in-India” initiative, cutting cost and lead times, securing INR 50M in orders and lifting portfolio margin 4.2% at a projected 5% CAGR
- Collaborated with the R&D team to integrate the ASCO Group 5 controller and MODBUS communications, enhancing system monitoring and control, ensured IEC-standards compliance.
- Led a team of 4 (external vendors and a product application expert) to build the Digital Sales Animation Tool, a 3D "Grid-to-Plug" tool spanning distribution networks and motor management, managing and working with the IT network team to build cybersecurity and offline operation.
- Built five-year product roadmaps and strategic plans, and managed demand forecasting, backlog and supply-chain planning across the transfer-switching portfolio.
- Co-created the Digital Power Train demonstration for ELECRAMA 2023, translating safety, availability, efficiency and sustainability propositions into a customer-facing concept.
- **Selected Recognition:** Innovation Pillar Award: Disruptive Product Launch, for localization of the ASCO S7000 IEC MV ATS, Schneider Electric, 2024; Three-time recognition for first-country order booking and exceeding launch targets by over 100% (TransferPacT), Schneider Electric, 2023-24.

Graduate Engineer Trainee
Schneider Electric · Bengaluru, India · October 2021 - August 2022

- Developed technical expertise across the transfer switching equipment portfolio, end-distribution networks, motor management and applicable IEC standards.
- Led the phase-wise launch of TransferPacT and WATSN, exceeding revenue targets by over 100% for both products.
- Created standards-aligned technical value propositions and executed go-to-market and post-launch demand-generation campaigns with sales and marketing teams.

EDUCATION

MSc, Electrical Power Systems Engineering

University of Manchester, United Kingdom · September 2026 · predicted Merit

Relevant modules: Power System Plant and Asset Management, Power System Operation and Economics, Power System Analysis and Control, Power System Dynamics and Stability, Power System Protection and Communication, Power System Planning and Resilience, Smart Distribution Networks, Electrical machines and Power Electronics

Dissertation: "Interaction of Environmentally Friendly (SF_6-free) Circuit Breakers with Power Transformers in a GIS": EMT modelling in PSCAD of switching transients and insulation-coordination interactions in gas-insulated switchgear for sustainable/net-zero HV substation design.

B. Tech, Electrical and Electronics Engineering

SRM Institute of Science and Technology, India · May 2021 · CGPA 8.00 / 10.00

Relevant modules: Electrical Machines, Power Electronics, Signals & Systems, Microcontroller Engineering, Power System Analysis, Power System Protection, Smart Grid, Hybrid Electric Vehicles Engineering.

Two peer-reviewed publications:

- **"Simulation of Circulating Current and Common Mode Voltage in Inverter-Fed Induction Motor Using PID and Fuzzy Controller,"** ***Solid State Technology*****, Vol. 63(3), 2020.**
  Designed a hybrid PID-Fuzzy controller in MATLAB/Simulink, validated on a dual-inverter hardware rig; reduced common-mode voltage by 94.6%, shaft voltage by 85.6% and circulating bearing current by 66.7%.
- **"Underground Cable Fault Detection Using Internet of Things,"** ***Cognitive Informatics and Soft Computing*** **(AISC vol. 1317), Springer, 2021.**
  Simulated and prototyped an Arduino/ESP8266 fault-location system in Proteus, achieving ±100 m location accuracy for underground distribution cables and reducing fault-rectification time.

KEY ENGINEERING PROJECTS

MSc Projects: University of Manchester (2025-26)

132 kV Double-Circuit Overhead Line: Design & Technoeconomic Analysis: designed an HV transmission line and ran a comparative analysis of conductor and route options against electrical-performance and cost criteria, applying design-code and clearance requirements.

Economic Dispatch to N-1 Security-Constrained Operation: modelled the cost-reliability trade-off from economic dispatch through to security-constrained (N-1) operation.

Overcurrent Protection & Relay Grading: performed protection coordination and time-current grading with supporting load-flow and voltage studies to size and verify safe network performance.

Queensland LV Network Planning under High Rooftop-PV Penetration: Led a team, modelling a real LV feeder to assess renewable hosting capacity; ranked reinforcement options by benefit-cost ratio; codebase in Python.

Power Flow & Renewable Inverter Integration: Implemented a BFS power-flow solver in MATLAB/Simulink for the CIGRÉ MV benchmark distribution network; a GFL inverter (PLL, dq current control, weak-grid testing)

INDUSTRIAL INTERNSHIPS

Lauritz Knudsen Electrical & Automation (2023): Selection, protection and application of LV switchgear and automation integration for industrial/commercial systems.

Bharat Heavy Electricals Ltd (BHEL, 2019): Hands-on exposure to 800–1200 kV transformer manufacturing (core assembly, winding, vacuum drying, oil impregnation) and HV testing (impulse, over-potential) in UHV labs.

NTPC (National Thermal Power Corporation,2019): Thermal generation processes, 500 MW turbines and high-capacity generators, generator protection for grid stability, and emissions/ash-utilization systems.

LEADERSHIP & VOLUNTEERING

Women & Allies Group (Engage Pillar), Schneider Electric (2024-2025): produced content promoting workplace gender equality and inclusion.

Panelist, Campus-to-Corporate Programme, Schneider Electric (2024): mentored new engineering trainees through onboarding panels and Q&A sessions.

Technical Team, EEE Association, SRM (2018-2021): organized technical workshops and events with industry experts across electrical subjects.

LANGUAGES

English (Professional Proficiency); Bengali (Native Proficiency); Hindi (Bilingual Proficiency); French (Basic Proficiency)